MILITARY SPECIFICATION

SOUND CHAMBERS AND CUFF, BLOOD PRESSURE, CONDUCTIVE

This specification has been approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of blood pressure sound chambers used with a conductive cuff.

1.2 Classification. The blood pressure sound chambers shall be of the following sizes:

Size	Sound Chambers
I	Adult
II	Medium adult

1.2.1 Cuff size. The conductive cuff shall be of one size for the two sizes of blood pressure sound chambers, namely, adult and medium adult.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard Shipping.
PPP-B-676 - Box, Set-Up.

MILITARY

MIL-C-37107 - Cuffs, Sphygmomanometer, Conductive.

STANDARDS

FEDERAL

FED. STD. NO. 601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

No. 24345 SOUND CHAMBERS AND CUFF, BLOOD PRESSURE, CONDUCTIVE.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
No. 56 Code for Use of Flammable Anesthetics.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, Mass. 02210).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Applications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

* 3.1.1 Sound chamber. The sound chamber shall be fabricated of Neoprene rubber of Shore Durometer A hardness of 46 ± 5 or Kraton elastomer of Shore Durometer A 39 ± 6 hardness when tested in accordance with 4.5.4. The short nipple shall be made of stainless steel or aluminum. The sound tubing shall be of food grade polyvinyl chloride plastic. A hook type strip of fastener shall be present on the top surface of the sound chamber.

3.1.2 Cuff. The blood pressure cuff shall be made of suitable conductive rubber-coated cloth. The fastening of the hook and pile type shall be non-sparking when tested in accordance with 4.5.3. The cuff shall comply with the NFPA No. 56 requirement for conductivity. The cuff shall withstand autoclaving in accordance with 4.5.1 without visible deterioration.

3.1.3 Compression bag. The bag shall be fabricated of latex rubber and coated with a suitable conductive rubber. The compression bag shall comply with the NFPA No. 56 requirement for conductivity.

3.2 Style, design and construction. The blood pressure sound chamber and cuff unit shall consist of a soft, neoprene rubber or Kraton elastomer contoured sound chamber with permanently attached metal nipple and attached vinyl sound tube of about 22 inches in length. The free end of the sound tube shall be fitted with a male Luer taper in order to be connected to a pair of aural tubes. The male Luer taper shall be in accordance with the ANSI Standard Z70.1 ring gage of figure 7. The top of the sound chamber shall have a strip of hook type fastening strip cemented to it. A complementary strip of pile fastener shall be attached to the sphygmomanometer cuff so that the sound chamber will remain in position during use.

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3.2.1 Cuff. The cuff includes the inflatable compression bag and is held in position around the patient's arm by means of hook and pile tapes appropriately situated and sewn onto the cuff. The style of the cuff shall be comparable to that of MIL-C-37107.

* 3.2.2 Compression bag. The compression bag shall be of seamless construction and shall have an inlet and outlet tube each of which shall be 8 ± 1 inches in length. The outlet tube shall be sealed with a removable plug.

3.2.3 Sound chambers. The sound chambers shall be in accordance with DPSC Drawing No. 24345.

3.3 Resistance to leakage of compression bag. The compression bag shall be capable of withstanding an internal pressure of 12 psig for one minute without leakage when tested according to 4.5.2.

3.4 Effect of autoclaving. There shall be no evidence of stiffening, softening or cracking of the rubber compression bag after autoclaving in accordance with 4.5.1. The requirement for resistance to leakage of the compression bag (see 3.3) shall be applied to the compression bag after it has been autoclaved as specified in 4.5.1.

3.5 Marking. The cuff and compression bag shall be legibly and permanently marked with the contractor's name or registered trademark. The cuff and compression bag shall be marked "CONDUCTIVE".

3.6 Workmanship. Workmanship shall be first class throughout. The complete unit shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling. The unit of product shall be for examination purposes, a sound chamber with accompanying cuff and compression bag.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent defective) 1/
Major	S-2	1.5
Minor	S-1	4.0

1/ If acceptance number is not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, inspection at Level II may be performed to determine final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II below:

TABLE II. Sampling criteria for tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL (percent defective)
Sound chamber	Hardness	3.1.1	4.5.4	S-1	2.5
Fastening strips	Non-sparking	3.1.2	4.5.3	S-1	1/
Cuff and compress-ion bag	Conductivity	3.1.2 &3.1.3	NFPA No. 56	S-1	1/

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TABLE II. Sampling criteria for tests.(cont'd)

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Luer taper fitting	Dimensions	3.2	ANSI Z70.1	S-1	2.5
Compression bag	Leakage	3.3	4.5.2	S-1	1/
Cuff, bag and sound chamber	Autoclaving	3.3 & 3.4	4.5.1	S-1	1/

1/ Acceptance number shall be zero for all sample sizes.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The sound chamber and cuff unit shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any sound chamber and cuff unit in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major 101	Surface of sound chamber in contact with arm is not smooth.
102	Fastener strip on cuff not positioned to hold sound chamber properly.
103	Compression bag not of seamless construction.
104	Inlet and outlet tubes of compression bag not smoothly conjoined with compression bag.
105	Inlet and outlet tubes shorter than required.

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TABLE III. Classification of defects. (cont'd)

Categories	Defects
Major	
106	Vinyl tubing kinked, i.e., restricting bore.
107	Fastener strips (hook and pile strips) not aligned for proper engagement.
Minor	
201	Name or trademark not present on cuff and compression bag.
202	Compression bag not stamped "CONDUCTIVE".

4.4 Packaging inspection. The inspection of the unit package, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Autoclaving. The cuff and compression bag shall be steam autoclaved at 29 to 31 psig and 250° F± 2° F for 30 minutes and then allowed to cool at room temperature (68° to 72° F) for one hour.

4.5.2 Resistance to leakage of compression bag. The compression bag shall be subjected to an internal pressure of 12 psig for one minute. The bag shall be secured from ballooning by use of screening of adequate strength.

4.5.3 Non-sparking test. The hook and pile fastener shall be locked together and then rapidly separated in a dark room (devoid of all lights) at a relative humidity of less than 35%. The interface shall not exhibit a blue streak or any other light reaction due to electrostatic build up. This procedure shall be repeated three more times without the production of visible light at the interface.

4.5.4 Hardness. The sound chamber shall be subjected to a Shore Durometer A hardness determination in accordance with Method 3021 of Fed. Test Method Std. No. 601.

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* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. Each sound chamber and cuff, of the type specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each sound chamber and cuff of the size specified shall be packaged in a polyethylene bag with the cuff and tubing arranged in such a manner that it will not kink or be permanently deformed. Excess air shall be expelled from the bag and closure shall be effected by heat sealing. Each item shall be further packaged in a box of appropriate size conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Twelve unit packages containing sound chamber and cuffs of one size only shall be packaged in a box constructed in accordance with PPP-B-636, type CF, class domestic. Closure of the box shall conform to method I as specified in PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Six intermediate packages containing sound chamber and cuffs of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Six intermediate containers containing sound chamber and cuffs of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Number and title of this specification.
b. Size (see 1.2).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).
d. Applicable National Stock Number.

6.2 This specification covers the following blood pressure sound chamber and cuff units that appear in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-420-5749	SOUND CHAMBER AND CUFF, Blood Pressure, Conductive, Adult
6515-00-435-8961	SOUND CHAMBER AND CUFF, Blood Pressure, Conductive, Medium Adult

It does not cover all types and sizes indicated by the title nor all types which are commercially available.

* 6.3 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-2008

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37123A SOUND CHAMBERS AND CUFF, BLOOD PRESSURE, CONDUCTIVE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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